

Analysis of Material Selection for Wind Turbine Blade Under Performance and Cost Constraints
Advait Nair

Abstract

This analysis develops a multi-criteria material selection model to evaluate structural materials for wind turbine blade applications under competing performance and economic constraints. Aluminum alloy, fiberglass composite, and carbon composite were compared using density, stiffness, fatigue resistance, and cost. Specific stiffness and specific strength were calculated to assess performance relative to weight, and all variables were normalized to a common scale. A weighted scoring framework was used to rank materials under performance-focused, balanced, and cost-focused scenarios. Results indicate that carbon composite provides superior structural performance due to its high stiffness-to-weight ratio and fatigue resistance, while fiberglass composite becomes more competitive under cost-sensitive conditions. The findings demonstrate that optimal material selection depends on design priorities rather than a single dominant property, while reaffirming why the status quo uses fiberglass composites for wind turbine blades.

Introduction

Wind turbine blades operate under complex structural and environmental conditions that impose strict material requirements. The blades on a wind turbine must be lightweight to reduce gravitational and rotational loads, stiff to prevent excessive deflection, and capable of withstanding repeated cyclic loading over long operational lifetimes. At the same time, they must remain economically feasible for large-scale energy deployment. As a result, material selection for wind turbine blades is determined by a balance of competing constraints rather than one specific variable.

In modern engineering practice, materials are rarely selected based solely on maximum strength or stiffness. Instead, engineers evaluate how efficiently a material delivers performance relative to its weight and cost. This is particularly important for wind turbine blades, where increased mass or material usage can increase structural stresses and reduce overall efficiency of the machine. Consequently, the problem of material selection becomes one of tradeoff analysis, where performance improvements must be weighed against economic and practical limitations.

This analysis develops a simplified but structured material selection model to evaluate three candidate materials: aluminum alloy, fiberglass composite, and carbon composite. The goal is not to produce an exact design solution, but to demonstrate how engineers can use normalized metrics and weighted decision to compare materials under varying priorities. By analyzing how rankings change between performance-focused and cost-focused scenarios, this analysis highlights the role of tradeoffs in real-world engineering decisions

Methodology

Material Selection and Property Definition

Three materials were selected to represent a range of structural and economic characteristics. Aluminum alloy was chosen as a lightweight metal baseline, fiberglass composite was selected as a commonly used structural composite, and carbon composite was included as a high-performance but costly alternative. These materials provide a meaningful comparison because they differ significantly in stiffness, density, fatigue resistance, and cost.

The model incorporates five primary properties: density, modulus, strength, fatigue score, and cost index. Density represents mass per unit volume and directly influences the weight of the blade. Modulus, also known as Young's modulus, represents stiffness and determines how much a material deforms and bends under load. Strength represents the maximum stress a material can withstand before failure. Fatigue score is used as a simplified representation of resistance to cyclic loading, which is critical for wind turbine blades that experience continuous fluctuating forces. Cost index represents relative material cost and is included to capture economic constraints. For the fatigue score, I used the fact that carbon-fiber composites are generally valued in wind blades because they have better stiffness, strength, and fatigue resistance per unit mass than glass-fiber composites, while fiberglass is still widely used and performs well, and aluminum is less attractive in this blade context. DOE/Sandia (Griffin et al., 2019) explicitly says carbon fiber has improved stiffness, strength, and fatigue resistance per unit mass, and that cost is the main reason it is not used everywhere.

The values used in this analysis are shown below.

Material Selection 4

materials <chr>	density <dbl>	modulus <dbl>	strength <dbl>	fat_score <dbl>	cost_index <dbl>	SpecStiff <dbl>	SpecStren <dbl>	WeightEff <dbl>
Aluminum Alloy	2700	69	300	4	5	0.02555556	0.1111111	0.0000000
Fiberglass Composite	1900	40	600	7	3	0.02105263	0.3157895	0.7272727
Carbon Composite	1600	100	1000	9	9	0.06250000	0.6250000	1.0000000

3 rows | 1-9 of 13 columns

fat_score <dbl>	cost_index <dbl>	SpecStiff <dbl>	SpecStren <dbl>	WeightEff <dbl>	CostEff <dbl>	SpecStiffEff <dbl>	FatigueEff <dbl>	TotalScore <dbl>
4	5	0.02555556	0.1111111	0.0000000	0.6666667	0.108642	0.0	0.1325926
7	3	0.02105263	0.3157895	0.7272727	1.0000000	0.000000	0.6	0.5245455
9	9	0.06250000	0.6250000	1.0000000	0.0000000	1.000000	1.0	0.8500000

3 rows | 5-13 of 13 columns

Derived Performance Metrics

To evaluate materials, two derived metrics were calculated: specific stiffness and specific strength. Specific stiffness is defined as modulus divided by density and represents how much stiffness a material provides per unit weight. Specific strength is defined as strength divided by density and represents how much strength is achieved per unit weight. By using these ratios, the model accounts for the fact that a material with high stiffness but excessive weight may not perform well in practice. Similarly, a material with moderate strength but low density may offer better overall efficiency.

Normalization and Scaling

Because the selected properties have different units and magnitudes, direct comparison is not meaningful without normalization. Each variable was scaled to a dimensionless range between 0 and 1. For properties where higher values are desirable, such as stiffness and fatigue resistance, normalization was performed so that the highest value approached 1 and the lowest approached 0. For properties where lower values are preferred, such as density and cost, normalization was reversed so that lower values corresponded to higher efficiency scores. This normalization process ensures that all variables contribute proportionally to the final score and allows for meaningful comparison across different categories.

Weighted Scoring Model

A weighted scoring model was used to combine the normalized variables into a single performance metric. The total score for each material was calculated as follows:

$$\text{Total Score} = 0.35(\text{Weight Efficiency}) + 0.30(\text{Stiffness Efficiency}) + 0.20(\text{Fatigue Efficiency}) + 0.15(\text{Cost Efficiency})$$

These weights were selected to reflect the importance of structural performance while still accounting for economic considerations. Weight and stiffness were prioritized because they directly influence blade deflection and structural loading, while fatigue and cost were included to capture long-term durability and feasibility. The model was implemented in R, where data was input, calculations were performed, and results were visualized.

The code used in this analysis is shown below.

3. Normalization

I cannot directly combine the variables because they are on different scales. So I turned them into scores between 0 and 1.

```
##For Higher is Good
## what this means is higher value is better for the operation of windmill i.e higher modulus means less bending so its better for windmill blades to have (because you don't want the blade to bend around)
norm_high_good <- function(x) {
  (x - min(x)) / (max(x) - min(x))
}

##For Lower is Good
## what this means is lower value is better for the operation of windmill i.e lower cost index means cheaper price so its better for windmill blades production
norm_low_good <- function(x) {
  (max(x) - x) / (max(x) - min(x))
}
```

4. Category Scores

```
## Weight efficiency | lower density is better
matsdf$WeightEff <- norm_low_good(matsdf$density)

## Cost efficiency | Lower cost index is better
matsdf$CostEff <- norm_low_good(matsdf$cost_index)

## Stiffness efficiency | higher specific stiffness is better
matsdf$SpecStiffEff <- norm_high_good(matsdf$SpecStiff)

## Fatigue efficiency | higher fatigue score is better
matsdf$FatigueEff <- norm_high_good(matsdf$fat_score)
```

5. Weightage of Each Category

```
w_weight <- 0.35
w_stiffness <- 0.30
w_fatigue <- 0.20
w_cost <- 0.15

## These are simple assumptions on how much a certain category affects the final score a material
## 35% of the final score comes from being lightweight
## 30% comes from stiffness
## 20% from fatigue
## 15% from cost
```

6. Total Score

```
#normalized values * weightage value added up
matsdf$TotalScore <- (w_weight*matsdf$WeightEff) + (w_stiffness*matsdf$SpecStiffEff) + (w_fatigue*matsdf$FatigueEff) + (w_cost*matsdf$CostEff)
```

Scenario-Based Analysis

To evaluate how design priorities influence material selection, 3 scenarios were defined. In the performance-focused scenario, there is greater emphasis placed on stiffness and weight efficiency. In the cost-focused scenario, cost is assigned a significantly higher weight. A balanced scenario (which is the original code) was also included to represent a more moderate design approach and juxtapose the other results. This scenario analysis allows the model to simulate engineering decisions, where priorities may shift depending on application and constraints of the project and environment.

The code used in this analysis is shown below.

Scenario Analysis

Previous model was running on the assumptions of weightage or how important certain categories are. What happens if I (the company) change my priorities? Maybe carbon is best only when performance matters most. Maybe fiberglass becomes more attractive when cost matters more. Balanced was already run above.

[Hide](#)

```
Scenario <- c("Performance_Focused", "Balanced", "Cost_Focused")

Weight_WeightEff <- c(0.35, 0.35, 0.25)
Weight_Stiffness <- c(0.35, 0.30, 0.20)
Weight_Fatigue <- c(0.20, 0.20, 0.15)
Weight_Cost <- c(0.10, 0.15, 0.40)

scenario_results <- data.frame()

for (i in 1:length(Scenario)) {
  temp_scores <- Weight_WeightEff[i] * matsdf$WeightEff +
    Weight_Stiffness[i] * matsdf$SpecStiffEff +
    Weight_Fatigue[i] * matsdf$FatigueEff +
    Weight_Cost[i] * matsdf$CostEff

  temp_df <- data.frame(
    Scenario = rep(Scenario[i], nrow(matsdf)),
    Material = matsdf$materials,
    Score = temp_scores
  )

  scenario_results <- rbind(scenario_results, temp_df)
}

scenario_results
```

Results

The results of the model reveal clear differences in material performance across scenarios. In the performance-focused scenario, carbon composite achieved the highest overall score. This result is driven by its superior specific stiffness and high fatigue resistance, which allow it to maintain structural integrity while minimizing weight. Its low density further enhances its performance in weight-sensitive applications such as wind turbine blades.

In the balanced scenario, carbon composite remained the highest-performing material, although fiberglass composite demonstrated a competitive performance too. Fiberglass benefits from moderate stiffness and significantly lower cost, allowing it to achieve a strong overall score despite lower performance metrics compared to carbon composite. Aluminum alloy consistently ranked lowest due to its higher density and relatively lower fatigue resistance, which reduced its effectiveness in cyclic loading environments.

In the cost-focused scenario, the material ranking shifted. Fiberglass composite became the most competitive and exceeded carbon composite in overall score. This shift is primarily due to the increased weighting of cost, which penalizes carbon composite heavily due to its high cost index. Although carbon composite maintains superiority in both structural integrity and performance, its economic disadvantage becomes the dominant factor when cost is prioritized.

Scenario<chr>	Material<chr>	Score<dbl>
Performance_Focused	Aluminum Alloy	0.1046914
Performance_Focused	Fiberglass Composite	0.4745455
Performance_Focused	Carbon Composite	0.9000000
Balanced	Aluminum Alloy	0.1325926
Balanced	Fiberglass Composite	0.5245455
Balanced	Carbon Composite	0.8500000
Cost_Focused	Aluminum Alloy	0.2883951
Cost_Focused	Fiberglass Composite	0.6718182
Cost_Focused	Carbon Composite	0.6000000
9 rows		

The greater the score, the preferable the material is used in the given context

Discussion

The results demonstrate that material selection for wind turbine blades is inherently dependent on the relative importance of performance and cost. Carbon composite provides the highest structural efficiency due to its high stiffness-to-weight ratio and strong fatigue resistance. However, its high cost limits its widespread adoption in cost-sensitive applications. This aligns with industry trends, where carbon fiber is used selectively rather than universally.

Fiberglass composite represents a balanced alternative that offers acceptable performance at a significantly lower cost. Although it does not match the stiffness or strength of carbon composite, it provides sufficient performance for many applications while maintaining economic feasibility. This

explains why fiberglass remains a widely used material in wind turbine blade manufacturing.

These findings highlight a key principle in engineering design: the best material is not necessarily the one with the highest performance, but the one that provides the best balance between constraints.

Limitations

This model is a simplified representation of material selection and does not capture the full complexity of real-world engineering. Composite materials were treated as isotropic with uniform properties, even though their behavior depends on fiber orientation, layup configuration, and manufacturing processes. Fatigue was represented using a simplified scoring system rather than a detailed fatigue-life model. Cost was modeled using a relative index rather than a comprehensive manufacturing and lifecycle cost analysis. As a result, the model can only be and should only be interpreted as a preliminary screening tool rather than a definitive design method.

Conclusion

The analysis demonstrates that multi-criteria decision modeling provides a useful framework for evaluating materials under competing constraints. Carbon composite offers superior performance in stiffness and fatigue resistance, while fiberglass composite provides a more cost-effective alternative. The results emphasize that material selection depends on design priorities and that tradeoffs play a central role in engineering decision-making. This approach can be extended to more complex models incorporating additional variables and more detailed data, such as real-time gathered data.

References

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